

Problem-Solution Fit

Date	25 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Small-to-medium scale farmers, agricultural researchers, agribusiness companies, farming policymakers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited technical literacy - Low budget for technology - Inconsistent internet access in rural areas - Basic smartphone/computer access	5. AVAILABLE SOLUTIONS PROS & CONS [AS] - Government agricultural offices (Pros: Free advice; Cons: Slow, generic) - Local farmer knowledge (Pros: Practical; Cons: Not data-driven) - Existing agri-apps (Pros: Digital; Cons: Complex, require internet, paid)
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Incorrect crop selection leading to low yield (Every Season) • Resource wastage on unsuitable crops (Frequent) • Lack of data-driven farming tools (Constant)	9. PROBLEM ROOT / CAUSE [RC] Farmers lack access to scientific analysis of their specific soil and environmental conditions; decision-making is based on intuition rather than data	7. BEHAVIOR + ITS INTENSITY [BE] - Manually selects crops based on tradition (High) - Consults neighbors and local dealers (Medium) - Occasionally watches farming videos online (Low)

2. TRIGGERS TO ACT [TR] • Start of a new farming season • Financial losses from previous crop failure • Government push for smart agriculture	10. YOUR SOLUTION [SL] OptiCrop — A web-based ML-powered crop recommendation system that takes soil (N, P, K, pH) and environmental parameters (temp, humidity, rainfall) as input and predicts the most suitable crop using trained machine learning models	8. CHANNELS OF BEHAVIOR [CH] ONLINE: YouTube farming videos, WhatsApp farming groups, agricultural websites
4. EMOTIONS BEFORE / AFTER [EM] Before: Uncertain, anxious, overwhelmed → After: Confident, informed, optimistic		OFFLINE: Local agricultural offices, farmer cooperatives, seed dealers